

Lab 3: Component Modelling & Architectural Pattern Selection

Hospital Bed & ICU Allocation Dashboard

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Architecture Selection

We chose the Layered Architecture (Presentation → Business → Data) for the Hospital Bed & ICU Allocation Dashboard System.

Reason 1: Clear Separation Between Real-Time Clinical Workflows and Data Integrity

The system serves two distinct actor groups — Triage Nurses who need a fast, glanceable view of the prioritized waitlist and bed status (FR-001, FR-002), and Hospital Administrators who must approve or reject sensitive transfer decisions (FR-003). A layered structure isolates the Dashboard UI Component from the business rules that compute triage severity, manage bed state transitions, and coordinate ambulance escalation (UC-01–UC-07). This means the triage-scoring algorithm or the bed-allocation workflow can be modified or extended (e.g., adding a new escalation rule) without touching the presentation code that nurses and administrators interact with directly.

Reason 2: Simplicity of Deployment for a Single-Facility Real-Time System

Unlike a multi-hospital or multi-tenant platform, this dashboard runs within one facility's ER, with a bounded, well-known set of use cases and a single authoritative source of bed/patient truth. A layered monolith avoids the operational overhead of a microservices architecture (independent deployment pipelines, inter-service network calls, distributed data consistency) while still keeping the Business Layer components (Triage & Waitlist Manager, Bed Allocation Manager, Notification Service, Ambulance Transfer Coordinator) modular and independently testable through well-defined interfaces such as `IBedTransferRequest` and `IAmbulanceTransfer`.

Security Advantage

Centralizing all patient, vitals, and transfer data access behind a single Data Layer (Patient/Bed Database and Audit & Transfer Log Store) lets the Auth & Access Control Service enforce authentication and role-based authorization (Triage Nurse vs. Hospital Administrator) at one consistent boundary, rather than duplicating access checks across many independent services. This directly supports NFR-002: every patient-data channel between layers can be encrypted in transit with TLS 1.2+, and all records at rest in the Data Layer can be encrypted with AES-256, since there is a single, well-defined data-access path (`IPatientDataAccess`, `IAuditWrite`) to secure and audit.

Performance Benefit

Because the Business Layer components sit in the same process/tier and communicate through in-memory calls rather than network hops between microservices, bed-status changes computed by the Bed Allocation Manager can be pushed by the Notification Service to the Dashboard UI Component over a persistent WebSocket connection with minimal added latency. This layered design makes it practical to meet NFR-001's requirement that bed status transitions propagate to all connected ER dashboard clients in under 1 second, since the update path (Bed Allocation Manager → Notification Service → Dashboard UI) avoids the queuing and serialization delays typical of cross-service network calls in a distributed microservices setup.

Trade-off Acknowledged

The layered approach does couple the Business Layer components to a single deployable unit, so scaling the Notification Service independently under extreme ER surge load is harder than it would be in a microservices design. For the scope of a single-facility ICU dashboard, however, this trade-off is acceptable given the gains in simplicity, consistency, and security.